GPS Renewables Pvt Ltd	CBG PROJECTS	ANNEXURE - 1

Inspection & Test Plan (ITP) For

Civil Construction Work

Rev	Date	Reason for Issue	Author	Checked	Approved
0	29/03/2024	Issued for review	Snehashish D/ Yeggireddi S	Amit Sharma / Trilok Kumar	Vithal Mane

		Rev
	QA/ICL/CIVIL/ITP/001	00

Revision History

Amendment Date	Revision Number	Amender	Amendment

Related documents

Document number	Document name	Description of Content

Holds

Section Number	Short Description of Hold

Definitions and Abbreviations

Term	Comment
IR	Inspection Record
TC	Test Certificate
LB	Logbook
SR	Site Register
TR	Test Report
W	Witness (The notified inspection activity may proceed if the involved parties does not attend)
R	Review (Verification of compliance with specifications by review of documentary evidence of QC related activities)
M	Monitoring (Random surveillance activity and review of document)
H	Hold point (Verification of compliance with specifications by witnessing activities and providing an Inspection record. The activity cannot proceed without witnessing)

	Inspection and Test Plan For Civil Construction Work				**Project**		**W**= Witness (The notified inspection activity may proceed if the involved parties does not attend)
	Client Name: IOCL	**TAG NO:**	**Document No.**	QA/ICL/CIVIL/ITP/001	**Package**		**R**= Review (Verification of compliance with specifications by review of documentary evidence of QC related activities)
		Description of Work:	**REV. No.**	0	**PO/WO No.**		**M**= Monitoring (Random surveillance activity and review of document)
	Item / Prodcut:		**DATE**	29/03/2024	**Construction Agency**		**H**= Hold point (Verification of compliance with specifications by witnessing activities and providing an Inspection record. The activity cannot proceed without witnessing)
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Inspection Stage nos	Activity	Characteristics/ Instruments	Type of Check	Quantum of Check	Reference Document & Acceptance Norms	Format of Record	Construction Agency	GPSR	Client	Remark/ Precaution
1	2	3	4	5	6	7	8	9	10	11
1	**GENERAL REQUIMENT**									
1.1	QA & QC laboratory. Availability of requisite laboratory set up and equipment in good working condition & valid calibration well before commencement of construction activities.	As agreed/required	Physical verification & review of calibration certificate	Before starting the construction activities and later on tracking calibration validity	Technical Spec and IFC drawings	SR/LB	H	H	M	Functioning & Calibration status of laboratory equipments in proper working condition to be verified after the lab setup and there after on random basis / check calibration validity
1.2	Sampling for testing of construction raw materials (Steel, Cement, Fine Aggregate, Coarse Aggregate)	As agreed/required	Physical	Once per each source	Technical Spec and IFC drawings	SR	H	M	R	Test Report from NABL accridiated laboratories to be submitted to Client for their review and acceptance.
1.3	Stacking and storage of construction materials and components at site.	As per IS : 4082	Physical	Random check per construction agency	Technical Spec and IFC drawing	SR	H	M	M	
2	**SURVEY**									
2.1	Excavation & Filling	Initial ground level before start of excavations. Shape, dimension, side slope & level of final excavation	Visual / Physical	Random Check	As per IFC drawing & Technical Spec	SR	H	H	R	Joint Protocol for co-ordinate and elevation
2.2	Suitability of filling material (Sand, GSB, WBM, Soil, Coarse Aggregate)	Grain size analysis, Organic matter, Liquid limit, Plastic limit,	Physical	Once per each type of source or during change of source	Tech spec and approved IFC drawing.	SR / TR	H	M	R	
2.3	Swell Index and chemical analysis (like Organic matter , Calcium carbonate , pH value , Total soluble Sulphate etc) as required (TS)	Lab test	Physical	minimum of 2 samples survey	As per technical spec	TC	H	H	R	LAB TEST
2.4	Back fillling soil - Compaction test (at lab & at Site)	Optimum moisture content (OMC) and max dry density (MOD) of filling / backfilling soil	Physical	As needed after back filling at site	Technical Spec and approved IFC drawings	SR/TR	H	M	R	After the preparation of soil bed if bacfilling done then compaction test may require to be done. Test is conducted by core cutting method, Rapid Moisture Meter or sand replacement test.
3	**MIX DESIGN**									
3.1	Sampling for lab testing of construction materials, (Course aggregate, Fine aggregate, Cement, Water, Admixture etc) materials for concrete mix design	As per lab guidelines & design requirement	Physical	Once per each new source or during the change	IS:456, IS:10262 & Tech spec	SR / LB / TR / Mix Design Report	H	H	M	Design Mix to be carried out at NABL accridiated Lab. Samples can be offered to the client before sending to the lab
3.2	Design Mix - Slump test, Cube samples, Sand, Cement, Aggregate testing at lab to be prepared as per IS:456 & compression strength to be checked	As per lab guidelines & design requirement	Physical	Once Prior to start of work	IS:516 & IS:456, IS:10262/ Technical Specifications	LB / TR / Mix Design Report	H	H	M	If samples doesn't meet the IS cretira then new Design Mix would be prepared for the Re design Mix
3.3	Trial Mix - Cubes samples at site as per Design Mix	As required for 7 & 28 Days strength test	Physical	Once Prior to start of work	IS:516 & IS:456, IS:10262/ Technical Specifications	LB / TR / Mix Design Report	H	H	**H**	Trial Mix Cube tests will be conducted at site after recieving the final Design Mix report from Lab Mixing shall be in GPS approved batching plant (at site or RMC Plant)

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4	**CONCRETING WORK**									
	COARSE AGGREGATE FOR CONCRETE									
4.1	Particle Size & Shape (Sieve Analysis, Flakiness Index, Elongation Index including visual Inspection)	Sieves (Confirming to IS:460-1962), Balance, Oven, Thickness gauge, Length Gauge, Metal Scoop etc	Physical	One test per 500 cu. M. /Change of Source, Whichever is earlier	IS:2386 Part-I (For Test Procedure), IS:383 (For Permissible Value) / TS	SR/LB	H	M	R	Results should be as per request of design mix subjected to verification within the limits specified in relevant code
4.2	Deteterious Materials & Organic Impurities	Balance, Sieve (Confirming to IS:460-1962)	Lab Analysis and Physical	Once per each new source or during the change	IS:2386 Part-II (For Test Procedure), IS:383 (For Permissible Value) / TS	Test Report	H	M	R	Results should be as per request of design mix subjected to verification within the limits specified in relevant code
4.3	Determination of Specific gravity, Water Absorbtion, Bulk density and Voids	Balance, Sieve (Confirming to IS:460-1962)	Physical	Once per each new source or during the change	IS:2386 Part-III (For Test Procedure), IS:383 (For Permissible Value) / TS	Test Report	H	M	R	Results should be as per request of design mix subjected to verification within the limits specified in relevant code
4.4	Determination of Crushing Value, Determination of Impact Value	Standard Appratus for these tests shall be used	Physical	Once per each new source or during the change	IS:2386 Part-IV (For Test Procedure), IS:383 (For Permissible Value) / TS	Test Report	H	M	R	Results should be as per request of design mix subjected to verification within the limits specified in relevant code
4.5	Moisture Content	Balance, Sieve (Confirming to IS:460-	Physical	Once per each new source or during the change	IS:2386 Part-III IS:456 IS:383/TS	Test Report	H	M	R	Results should be as per request of design mix subjected to verification within the limits specified in relevant code
4.6	Soundness	Reagents (Sodium Sulphate or Magnesium Sulphate)	Chemical/ Physical	Once per each new source or during the change	IS:2386 Part-V (For Test Procedure), IS:383 (For Permissible Value) / TS	Test Report	H	M	R	Results should be as per request of design mix subjected to verification within the limits specified in relevant code
5	**FINE AGGREGATE**									
5.1	Bulk Density, Surface Moisture & water Absorption	As per IS Code	Physical	Once per each new source or during the change	IS:2386 Part-III IS:383/ TS	Test Report	H	R	R	Results should be as per request of design mix subjected to verification within the limits specified in relevant code
5.2	Mortar making properties	As per IS Code	Physical	Once per each new source or during the change	IS:2386 Part-VI IS:383/TS	Test Report	H	R	R	Results should be as per request of design mix subjected to verification within the limits specified in relevant code
5.3	Silt, Clay Content and organic impurities and deleterious materials	Balance, Sieve (Confirming to IS:460-1962)	Physical	Once per each new source or during the change	IS:2386 Part-II (For Test Procedure), IS:383 (For Permissible Value) / TS	Test Report	H	M	R	Results should be as per request of design mix subjected to verification within the limits specified in relevant code
5.4	Sieve Analysis, Particle Shape & Size	Set of Sieves	Physical	Once per each new source or during the change	IS:2386 Part-I IS:383/TS	Test Report	H	R	R	Results should be as per request of design mix subjected to verification within the limits specified in relevant code
5.5	Soundness	Reagents (Sodium Sulphate or Magnesium Sulphate)	Chemical/ Physical	Once per each new source or during the change	IS:2386 Part-V (For Test Procedure), IS:383 (For Permissible Value)	Test Report	H	R	R	Results should be as per request of design mix subjected to verification within the limits specified in relevant code
6	**WATER**									
6.1	Water lab test	TDS, PPM, pH	Testing at Lab	One per three month for each source	IS:3025 IS:456	SR/LB/TR	H	R	R	These tests shall be carried out while establishing design mix & results be intimated and every three month water test is required.
7	**CEMENT-OPC43 Grade**									
7.1	Ensure that cement is stored in weather tight covered storage on raised platdorm	Preservation	Visual	100% covered storage	Standard preservation practices	SR/LB/TR	H	R	R	Manufacturer test certificate to be submitted by the contractor. & Manufacturing Date to be checked at site,
7.2	fineness	As required	Physical	Once per each Brand or during the change	IS:4031/1489/269/455/TS	SR/LB/TR	H	R	R	Manufacturer test certificate to be submitted by the contractor.
7.3	Initial & Final setting time	As required	Physical	Once per each Brand or during the change	IS:4031/TS	SR/LB/TR	H	R	R	Manufacturer test certificate to be submitted by the contractor.
7.4	Soundness	As required	Physical	Once per each Brand or during the change	IS:4031/TS	SR/LB/TR	H	R	R	Manufacturer test certificate to be submitted by the contractor.
7.5	Specific Gravity	As required	Physical	Once per each Brand or during the change	IS:4031/TS	SR/LB/TR	H	R	R	Manufacturer test certificate to be submitted by the contractor.

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7.6	Compressive Strength	As required	Physical	Once per each Brand or during the change	IS:4031/TS	SR/LB/TR	H	R	R	Manufacturer test certificate to be submitted by the contractor.
8	**FOUNDATION SYSTEM**									
8.1	Foundation casting/layout/shape/RCC work/curing etc.	As per reuired	Physical	Each Foundation	As per Tech spec and approved IFC drawing	SR / LB / IR	H	W	R	Alignment as per the coordinates of approved IFC drawing to be checked. Concrete grade to be checked as per design mix
9	**STAGING, SHUTTERING AND FORMS**									
9.1	Form Line, level and dimensions as per drawing, free from sap, shakes, wormholes, warps or other surface defects & having smooth finish, Adequate cross bracing of supporting framework.	As required	Visual/ Physical	During the entire Construction Phase	As per approved drawings	SR/LB	H	M	**M**	Length, Breadth, Depth, Shape, Level, Plumb line to be checked. Diagonal bracings are to be checked
10	**CONCRETE PLACING & COMPACTION**									
10.1	Batching Plant/Source of Concrete	Mixing of concrete shall be done in a batching plant such as to produce a homogeneous mix	Physical	To be calibrated at the time of starting and every 3 months	Review of calibration chart/certificate, IS:456/TS	SR / LB	H	M	R	Time of mixing will be as given in technical specification/IS 456. Minimum time of mixing should be 2 minutes for mixer capacity 2 CuM or less. Minimum time of mixing should be 3 minutes or as recommended by the mixer manufacturer for mixer capacity above 2 CuM.
10.2	Verification of Concrete Batch report	Concrete material proportion as per design mix	Physical	At the receipt of every new concrete batch	Design mix report	SR	H	M	R	Concrete mix batch reports / slips records to be maintained
10.3	Handling and Conveying	Buckets, chutes	Physical	100%	As per Technical Specification, Construction/Erection	SR	H	R	R	Technical Specification is to be followed.
10.4	Placement of Concrete (as per pour card)	Visual	Physical	100%	As per Technical Specification, Construction/Erection	SR / Pour Card	H	H	M	No concrete shall be placed until the place of deposit has been thoroughly inspected and accepted & pour card is verified and signed
10.5	Compacting	As required	Physical	100%	Check for segregation as per IS:456 & Technical Specification	SR	H	R	R	Bleeding or segregation etc to be avoided as far as possible. Accumulated Water is to be removed and shall not be covered with concrete or dry concrete
10.6	Curing	As required - Minimum 10 days with water	Physical	100%	Check for period of curing as per IS:456 & Technical Specification	SR	H	R	R	Exposed concrete surface shall be protected against heating and drying for atleast 72 hours after placement for water curing method.
11	**PRE CONCRETE / DURING CASTING**									
11.1	Workabilty:- Slump test	Standard apparatus for different method used for measuring workability, slump cone	Physical	At the receipt of new concrete batch from every mixing plant/IS:456	IS:456, IS:1199-1959 & GPS Technical Specification	SR/LB/Test Report	H	M	R	Slump Test for medium and high workability, As per Design Mix Report
11.2	Cube sampling test	Concrete cube strength 7 days and 28 test	physical	as per IS:456	IS : 456 and Tech spec.	SR / LB / TR	H	M	R	Min of 6 cubes for each mix. Test 3 cube samples after 7days & remaining 3 cube samples after 28 days for compression strength
11.3	Water Cement Ratio	As required	Physical / verification of Batch report	At random at the time of batching	As per IS:1199 and approved design mix	SR/LB	H	R	R	Concrete batch report values to be verified against the Design mix parameters
11.4	Check for Cement Content	As required/As per design mix	Physical	At random at the time of batching	As per IS:3026, IS:456, approved design mix & technical specification	SR/LB	H	R	R	Concrete batch report values to be verified against the Design mix parameters
11.5	Admixtures for Concrete (If any)	As per IS:9103	Visual / Physical	At the receipt of new concrete batch from every mixing plant	IS:456, approved design mix &	Test Report	H	R	R	Admixture of approved brand and tested quality shall be used. Manufacture's TC required along with concrete batch report
11.6	Visual examination of finished structure	As required	Visual	100%	As per Technical Specification/	Test Report	H	M	M	

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11.7	Dimensions	As required	Measurement	100%	As per Technical Specification/	Test Report	H	H	M	
12	**POST CONCRETE**									
12.1	Visual Inspection of concrete surface just after removal of shuttering	As require	Visual	100%	As per IS : 456 / Technical specification	SR	H	H	M	
12.2	Dimension check on finished structures.	As require	Measure ment	100%	As per IS : 456 / Technical specification / Approved IFC drawings	SR/LB	H	H	M	
12.3	Position and alignment of Embedded parts and inserts	As require	Visual	100%	As per provision and tolerances of equipment supplier, Tech. spec and approved IFC drawings.	SR	H	H	M	
12.4	Submission of grouting / repair methodology to EIC for approval if concrete surface position and alignment of embedded parts / inserts are found defective.	As require	Review and approve	Once for each type of defect.	As per provisions and tolerances, Tech spec and IFC drawings.	SR	H	H	M	
12.5	Ultrasonic Pulse Velocity Test	For Critical Foundations	NDT	Based on the need	IS:13311 Part-I and Technical Specifications/IS:456	Test Report	H	H	M	This test is done if there is a failure in compression strength on sample cubes (after 28 days). UT shall be performed by NDT qualified person / institution with calibrated equipment in presence of GPS Engineer.
12.6	Load Test, Core Test and Rebound Hammer	As required	Test	Based on the need	Technical Specifications/ IS:456-2000	Test Report	H	H	**H**	These tests are done if there is a failure in compression strength on sample cubes (after 28 days). These tests also can be carried out in case of doubt regarding grade & quality of concrete
13	**REINFORCEMENT STEEL - TMT FE 500D**									
13.1	Physical and Chemical Properties as per relevant IS codes	As required/agreed	Review of MTCs	Every lot	IS 1786/456/TS	MTC	H	R	R	
13.2	Visual inspection to verify defect free reinforcement steel	As required/agreed	Visual	Random before and during placement	As per technical specification & IS: 468	SR	H	R	R	Any of the bars selected for use shall be free from cracks, surface flaws,laminations and rough, jagged and imperfect edges. Steel issued should be free from excessive rust. To be stored diameter wise in such a place so as to permit easy approach for inspection & identification. It should be cleaned of excessive rust before use.R/f will be tied with annealed wire. Water accumulation & distortion of r/f are to be avoided.
13.3	Placement of reinforcemnt steel	As required/agreed	Visual	100%	As per approved drawings and approved Bar Bending schedule/IS 456/TS	SR/LB Pour card	H	H	M	Check as per Bar Bending schedule (BBS) and as per relevent codal provision
14	**GROUT**									
14.1	Grouting Pressure	Calibrated Pressure	Physical	At Random	Approved drawing	SR/LB	H	H	**H**	
14.2	Composition of Grout	Nil	Verification of MTC	Each Lot/Batch	As per Technical Specification	SR/LB	H	H	M	

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15	BRICK MASONARY, SHEETING & ALLIED WORKS									
	TEST ON BRICKS									
15.1	Soundness	As Required	Physical	Once per each new source or during the change	IS:1077, IS:3495, IS:2691, Technical Specification	SR/LB	H	M	R	To be checked at site as per frequency stipulated in specification & guideline
15.2	Color	As Required	Visual	One for each Stack	IS:1077, IS:3495, IS:2691, Technical Specification	SR/LB	H	M	R	To be checked at site as per agreed guideline
15.3	Compressive Strength	As Required	Physical	As per relevant IS Code/ One Sample for 10000 Nos of lot	IS:1077, IS:3495 (Part I), IS:2891, Technical Specification	SR/LB	H	M	R	To be checked at site as per frequency stipulated in specification & guideline
15.4	Water Absorption	As Required	Physical	As per relevant IS Code/ One Sample for 10000 Nos of lot	IS:1077, IS:3496 (Part I), IS:2891, Technical Specification	SR/LB	H	M	R	To be checked at site as per frequency stipulated in specification & guideline
15.5	Visual & Dimension	Visual & Measurement Tape	Visual & Measurement	Random	IS:1077, Technical Specification	SR/LB	H	M	R	To be checked at site as per frequency stipulated in specification & guideline
15.6	Warpage	Straight Edge & Scale	Measurement	As per relevant IS Code/ One Sample for 10000 Nos of lot	IS:1077	SR/LB	H	M	R	To be checked at site as per frequency stipulated in specification & guideline
16	SAND (MASONARY WORK)									
16.1	Silt, Clay Content and organic impurities	Visual	Physical	Once per source & one for every change of source	IS:2386 Part-II, IS:383/Technical Specification	SR/LB/Test Report	H	M	R	At Site before start Masonary Works
16.2	Grading of Sand	IS Sieves	Physical	Once per source & one for every change of source	IS:2116/IS:383/Technical Specification	SR/LB/Test Report	H	M	R	At Site before start Masonary Works
17	JOINTS IN CONCRETE/DAMP PROOF COURSE									
17.1	Joint material —Bitumen impregnated fiber board, PVC water stops, sealing compound. Expanded polystyrene board, Hydrophilic strip, Acrylic polymer etc.(As given in Tech.specs.)	As per manufacturer standards	Review of MTC / TR	Each batch / Lot of Delivery	As per Tech spec and Construction drawings , IS:1838,1834 & IS:12200	SR / MTC	H	R	R	
17.2	Damp Proof Course	Material - Hot Bitumen and water proofing materials etc.(as given in Tech specs.)	Review of MTC / TR	Each batch / Lot of Delivery	As per Tech spec and IFC drawings.	SR / MTC	H	R	R	
17.3	Primer Thickness	Elcometer	Measurement	Each erection Mark	Technical specs and IFC drawings.	SR	H	M	R	
17.4	Final Paint inspection	Acceptance of painted surface, Finish, Shade	Physical	Random per each construction location	Technical specs and Const.drawings.	SR	H	M	R	
17.5	Stacking and storage of painting material	Material : Profiled colour coated metal deck & Cladding sheet	Review of MTC / TR	As per IS : 4082 / On the receipt of each new lot of material	Technical specs and Const.drawings / Profiled drawing.	MTC / TR	H	M	R	
18	DOORS									
18.1	Steel Doors	Materials & Check for shape, tolerances, Thickness, welding, finishing of sections as per Tech.specs.	Visual / Physical / TR	For each lot received at site	Technical specs and Const.drawings.	SR / LB / TR	H	M	R	MTC shall contain all the parameters specified in the technical specifications & as require per IFC drawings

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19	FENCING AND GATES									
19.1	Sampling for testing of Fencing material	Materials	Review of MTC / TR	Once per each source	Tech Specs and IFC drawings.	SR / MTC	H	R	R	MTC shall contain all the parameters specified in the technical specifications & as require per IFC drawings
20	FLOOR FINISHES AND ALIED WORKS									
20.1	Cement concrete flooring (PCC - Plain Cement Concrete)	Visual and dimension	Physical	Random during each PCC flooring work	Technical specs and IFC drawings	SR	H	R	R	
20.2	Ceramic tiles, Vitrified tiles, Glass, Mosaic, acid alkali resistant, heavy duty cement concrete	Visual & Material Properties	Review of MTC / TR	Each lot of delivery	Technical specs and IFC drawings	SR / MTC	H	R	R	MTC shall contain all the parameters specified in the technical specifications.
21	ANTI TERMITE TREATMENT DURING FLOORING									
21.1	Anti termite material	Material properties	Review of MTC / TR	Each lot of delivery	Tech spec / IFC Drawings/As per IS CODE 6313 Part 02	SR / TR / MTC	H	R	R	Before applying anti termite ensure material meets the requirement as per IFC material / IS codes/ MTC
21.2	Anti termite treatment inspection during flooring	Execution of treatment	Physical	Random check for each treatment	Tech spec / IFC Drawings/As per IS CODE 6313 Part 02	SR	H	R	R	Before applying anti termite ensure material meets the requirement as per IFC material / IS codes/ MTC
22	BITUMINOUS ROAD									
22.1	**GSB - Granular Sub base** (Sieve/Gradation, CBR, Max. dry density , Aggregate impact value, Moisture content prior to compaction)	Material properties	Chemical/ Physical	Once per each source	As per Tech spec and Construction drawings , IS:2720 & IS 2386	LB/Test Report	H	R	R	
22.2	**WBM - Water bond Macadam** (Sieve/Gradation, Max. dry density &OMC, Field Density compaction, Water absorption)	Material properties	Chemical/ Physical	Once per each source	As per Tech spec and Construction drawings , IS 2386 & IS 2720	LB/Test Report	H	R	R	
22.3	**DBM - Dense Bituminous Macadam** {Quality of binder , Soundness, water absorption of aggregate, Plasticity index (fine aggregate)}	Material properties	Chemical/ Physical	Once per each source	As per Tech spec and Construction drawings , IS:73, IS 2386 & IS 2720	LB/Test Report	H	R	R	
22.4	**BC - Bituminous Concrete** {Quality of binder , Soundness, water absorption of aggregate, Plasticity index (fine aggregate), Flakiness test}	Material properties	Chemical/ Physical	Once per each source	As per Tech spec and Construction drawings , IS:73, IS 2386 & IS 2720	LB/Test Report	H	H	**M**	
23	CEMENTIOUS WATERPROOFING									
23.1	Application as per procedure submitted by the supplier/Applicator	Visual inspection	Visual	Randam check during application	Tech spec / IFC Drawings	SR/LB	H	H	**H**	As per standard procedure of Brand
24	SLIPFORM SHUTTERING									
24.1	Fixing of Formwork, Shutterings Plates, Jacks, Alingment, Line Dia, etc	Visual inspection	Visual	Randam check during application	Tech spec / IFC Drawings	SR/LB	H	H	**H**	As per standard procedure of Brand